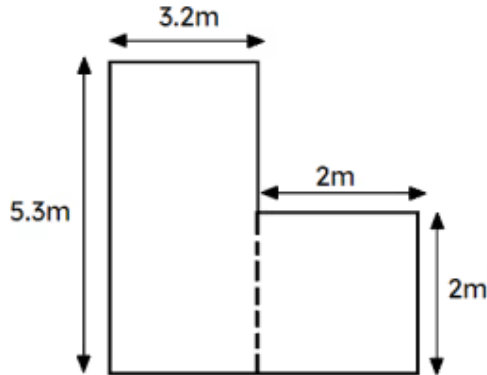




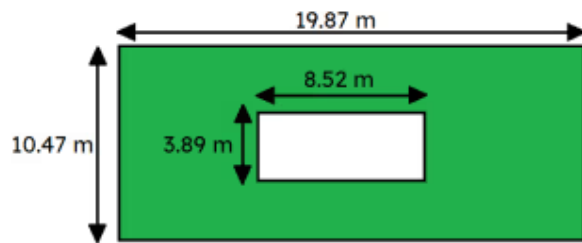
Problem solving with estimation and rounding

1 Work out an estimate for the area of the shape. Tick **1** correct answer



- ☐ 5 m²
☐ 10 m²
☐ 12 m²
☐ 19 m²
☐ 25 m²

2 Here is a grassed garden with a summer house in the centre. Using estimation and rounding, estimate the area of the grass. Tick **1** correct answer



- ☐ 50 m²
☐ 90 m²
☐ 100 m²
☐ 164 m²
☐ 800 m²

3 A piece of string was 32 cm measured to the nearest cm. A length of 5.6 cm (to 1 d.p) was cut off. What is the error interval for the remaining length x cm? Tick **1** correct answer

- ☐ $25.85 < x < 26.95$
☐ $25.85 \leq x < 26.95$
☐ $25.95 < x < 26.85$
☐ $25.95 \leq x < 26.85$



4 Complete the statement. Fermi problems encourage: Tick **3** correct answers

- ☐ critical thinking and well calculated estimates using data or information
- ☐ openness of answers and flexibility so answers can be achieved in different ways
- ☐ an appreciation there are a range of answers, the focus is on the working out
- ☐ random guesses but use estimation and or rounding

5 Fermi problem : How long will it take to count to 1 million (without breaks, sleep, etc)?
Tick **1** correct answer

- ☐ 50 days
- ☐ 40 days
- ☐ 16 days
- ☐ 5 days
- ☐ 3 days

6 Fermi problem : If you put 1 million pound coins in a box, what is the volume of the box in m^3 ? Tick **1** correct answer

- ☐ 0.5 m^3
- ☐ 1.2 m^3
- ☐ 5 m^3
- ☐ 8 m^3
- ☐ 12 m^3

